

Sadeep Ariyarathna

Systems Engineer | Mechatronics Engineer

Personal Information

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Experience

10/2024 – 08/2025	CLAAS E-Systems, Dissen am Teutoburger Wald, Germany Master Student - Modelling system architecture in Enterprise Architect, focusing on system engineering concepts. - Writing master thesis on collaboration techniques for merging and reviewing architecture models.
09/2022 – 02/2023	AtlasLabs (Pvt.) Ltd, Sri Lanka Mechatronics Engineer - Engineered and tested mechanical components and systems from concept to launch with Fusion 360, reducing product development cycle time by 20%. - Created manufacturable designs using Fusion 360 and validated them for Design for Manufacturing and Design for Assembly techniques. - Reduced troubleshooting time by 35% by creating bills of materials and validation documents for inspections.
07/2021 – 02/2022	MAS Intimates (Pvt) Ltd Automation Intern - Created SolidWorks designs using Design for Manufacturing and Design for Assembly techniques. - Designed and developed an attachment for sewing machines that reduced the time taken to cut the finished piece by 65%. - Developed control systems for three ongoing projects to evaluate each conceptual method, using microcontrollers (Arduino, ATTiny84).

Education

04/2023 - Present	Fachhochschule Südwestfalen, Soest - MSc. in Systems Engineering and Engineering Management (GPA – 1.9) - Thesis: A Collaboration Workflow for Merging System Architecture Models - An Example from Agriculture Industry (Writing)
2022	General Sir John Kotelawala Defence University, Sri Lanka - BSc. (Hons.) in Mechatronics Engineering (GPA – 1.8) - Thesis: Development of Elbow Prosthetic Control System

Projects

Chess Game	- Developed a fully functional chess game with time management, move validation, move history and move notation features. - Developed a reinforcement learning algorithm using DQN to train a chess agent.
IEEE BibTex Generator	- Developed a Python package to simplify the creation of BibTex entries for sources used in Latex. - The package can automatically generate the IEEE BibTex entry for a given DOI.
Remote Temperature Detection and Transmit	- Developed an application to detect temperature at a remote location and send using a GSM module. - The application was developed for STM32 platform using C. - The application has the capability to store the temperature readings and transmits using the GSM module via MQTT.
Grid Solving with Reinforcement Learning	- Developed a custom grid environment and analysing performance of Q-learning, Deep Q-learning and A2C algorithms using Pytorch, OpenAI Gym and StableBaseline3.
Prosthetic Elbow Concept	- Engineered a C++-based control system for precise control of an elbow joint using a brushless DC motor within a prosthetic arm. - Designed and optimized the intricate gear system for the joint utilizing SolidWorks, ensuring seamless integration and functionality.

Technical Skills

Programming Languages	Python, C/C++, Arduino, MATLAB
Software	Enterprise Architect, SolidWorks, Fusion 360, MATLAB Simulink, Autodesk Eagle, CubeIDE
Developer Tools	Git, Azure DevOps, Docker, Jenkins, Flask, Terraform, Ansible
Libraries	Pytorch, Tensorflow, StableBaseline, OpenAI Gym

Other Skills

IT	Microsoft Office, Adobe Photoshop, HTML & CSS
Hands-on-Skills	Handling Workshop tools, Soldering, PCB Designing

Languages

German	Conversational (Learning)
English	Fluent

Hobbies

- Playing Guitar
- Playing Computer Games